MSG No. 74.562 I_bmma [Type IV, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `mmm'`

No.	position	mapping
1	$\left[0, \frac{3}{4}, 0\right]$	`[1,4,6,7,18,19,21,24]`
2	$\left[0, \frac{1}{4}, 0\right]$	`[2,3,5,8,17,20,22,23]`
3	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{2}\right]$	`[9,12,14,15,26,27,29,32]`
4	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{2}\right]$	`[10,11,13,16,25,28,30,31]`

Table 2: Wyckoff site: `4b`, site symmetry: `mmm'`

No.	position	mapping
1	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	`[1,4,6,7,18,19,21,24]`
2	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	`[2,3,5,8,17,20,22,23]`
3	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	`[9,12,14,15,26,27,29,32]`
4	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	`[10,11,13,16,25,28,30,31]`

Table 3: Wyckoff site: `4c`, site symmetry: `mm'm'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,5,6,19,20,23,24]`
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,4,7,8,17,18,21,22]`
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,10,13,14,27,28,31,32]`
4	$\left[\frac{1}{2}, 0, 0\right]$	`[11,12,15,16,25,26,29,30]`

Table 4: Wyckoff site: `4d`, site symmetry: `mm'm'`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,5,6,19,20,23,24]`
2	$\left[0, \frac{1}{2}, 0\right]$	`[3,4,7,8,17,18,21,22]`
3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,10,13,14,27,28,31,32]`
4	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[11,12,15,16,25,26,29,30]`

Table 5: Wyckoff site: `8e`, site symmetry: `.2/m.`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[1,7,11,13]`
2	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[2,8,12,14]`
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[3,5,9,15]`
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[4,6,10,16]`
5	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[17,23,27,29]`
6	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[18,24,28,30]`
7	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[19,21,25,31]`

8	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[20,22,26,32]

Table 6: Wyckoff site: 8f, site symmetry: .2/m'.

No.	position	mapping
1	$\left[\frac{1}{4}, 0, \frac{1}{4}\right]$	[1,11,23,29]
2	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	[2,12,24,30]
3	$\left[\frac{3}{4}, \frac{1}{2}, \frac{3}{4}\right]$	[3,9,21,31]
4	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[4,10,22,32]
5	$\left[\frac{3}{4}, 0, \frac{3}{4}\right]$	[5,15,19,25]
6	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	[6,16,20,26]
7	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[7,13,17,27]
8	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	[8,14,18,28]

Table 7: Wyckoff site: 8g, site symmetry: mm2

No.	position	mapping
1	$\left[0, \frac{3}{4}, z\right]$	[1,4,6,7]
2	$\left[0, \frac{1}{4}, -z\right]$	[2,3,5,8]
3	$\left[\frac{1}{2}, \frac{1}{4}, z+\frac{1}{2}\right]$	[9,12,14,15]
4	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{2}-z\right]$	[10,11,13,16]
5	$\left[0, \frac{1}{4}, z\right]$	[17,20,22,23]
6	$\left[0, \frac{3}{4}, -z\right]$	[18,19,21,24]
7	$\left[\frac{1}{2}, \frac{3}{4}, z+\frac{1}{2}\right]$	[25,28,30,31]
8	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{2}-z\right]$	[26,27,29,32]

Table 8: Wyckoff site: 8h, site symmetry: mm'2'

No.	position	mapping
1	$[0, 0, z]$	[1,6,20,23]
2	$[0, 0, -z]$	[2,5,19,24]
3	$\left[0, \frac{1}{2}, -z\right]$	[3,8,18,21]
4	$\left[0, \frac{1}{2}, z\right]$	[4,7,17,22]
5	$\left[\frac{1}{2}, \frac{1}{2}, z+\frac{1}{2}\right]$	[9,14,28,31]
6	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-z\right]$	[10,13,27,32]
7	$\left[\frac{1}{2}, 0, \frac{1}{2}-z\right]$	[11,16,26,29]
8	$\left[\frac{1}{2}, 0, z+\frac{1}{2}\right]$	[12,15,25,30]

Table 9: Wyckoff site: 8i, site symmetry: 2'mm'

No.	position	mapping
1	$\left[x, \frac{3}{4}, 0\right]$	[1,7,18,24]
2	$\left[x, \frac{1}{4}, 0\right]$	[2,8,17,23]
3	$\left[-x, \frac{1}{4}, 0\right]$	[3,5,20,22]

continued ...

Table 9

No.	position	mapping
4	$\left[-x,\ \frac{3}{4},\ 0\right]$	[4,6,19,21]
5	$\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{1}{2}\right]$	[9,15,26,32]
6	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{2}\right]$	[10,16,25,31]
7	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{1}{2}\right]$	[11,13,28,30]
8	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{2}\right]$	[12,14,27,29]

Table 10: Wyckoff site: 8j, site symmetry: 2m'm'

No.	position	mapping
1	$[x,\ 0,\ 0]$	[1,2,23,24]
2	$\left[-x,\ \frac{1}{2},\ 0\right]$	[3,4,21,22]
3	$[-x,\ 0,\ 0]$	[5,6,19,20]
4	$\left[x,\ \frac{1}{2},\ 0\right]$	[7,8,17,18]
5	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[9,10,31,32]
6	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	[11,12,29,30]
7	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[13,14,27,28]
8	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	[15,16,25,26]

Table 11: Wyckoff site: 8k, site symmetry: m2'm'

No.	position	mapping
1	$[0,\ y,\ 0]$	[1,6,19,24]
2	$[0,\ -y,\ 0]$	[2,5,20,23]
3	$\left[0,\ y+\frac{1}{2},\ 0\right]$	[3,8,17,22]
4	$\left[0,\ \frac{1}{2}-y,\ 0\right]$	[4,7,18,21]
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[9,14,27,32]
6	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[10,13,28,31]
7	$\left[\frac{1}{2},\ y,\ \frac{1}{2}\right]$	[11,16,25,30]
8	$\left[\frac{1}{2},\ -y,\ \frac{1}{2}\right]$	[12,15,26,29]

Table 12: Wyckoff site: 8l, site symmetry: m2'm'

No.	position	mapping
1	$\left[\frac{1}{2},\ y,\ 0\right]$	[1,6,19,24]
2	$\left[\frac{1}{2},\ -y,\ 0\right]$	[2,5,20,23]
3	$\left[\frac{1}{2},\ y+\frac{1}{2},\ 0\right]$	[3,8,17,22]
4	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	[4,7,18,21]
5	$\left[0,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[9,14,27,32]
6	$\left[0,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[10,13,28,31]
7	$\left[0,\ y,\ \frac{1}{2}\right]$	[11,16,25,30]
8	$\left[0,\ -y,\ \frac{1}{2}\right]$	[12,15,26,29]

Table 13: Wyckoff site: `16m`, site symmetry: `.2.`

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ \frac{1}{4}\right]$	`[1,11]`
2	$\left[\frac{1}{4},\ -y,\ \frac{3}{4}\right]$	`[2,12]`
3	$\left[\frac{3}{4},\ y+\frac{1}{2},\ \frac{3}{4}\right]$	`[3,9]`
4	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	`[4,10]`
5	$\left[\frac{3}{4},\ -y,\ \frac{3}{4}\right]$	`[5,15]`
6	$\left[\frac{3}{4},\ y,\ \frac{1}{4}\right]$	`[6,16]`
7	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	`[7,13]`
8	$\left[\frac{1}{4},\ y+\frac{1}{2},\ \frac{3}{4}\right]$	`[8,14]`
9	$\left[\frac{1}{4},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[17,27]`
10	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[18,28]`
11	$\left[\frac{3}{4},\ y,\ \frac{3}{4}\right]$	`[19,25]`
12	$\left[\frac{3}{4},\ -y,\ \frac{1}{4}\right]$	`[20,26]`
13	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[21,31]`
14	$\left[\frac{3}{4},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[22,32]`
15	$\left[\frac{1}{4},\ -y,\ \frac{1}{4}\right]$	`[23,29]`
16	$\left[\frac{1}{4},\ y,\ \frac{3}{4}\right]$	`[24,30]`

Table 14: Wyckoff site: `16n`, site symmetry: `..m'`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,24]`
2	$[x,\ -y,\ 0]$	`[2,23]`
3	$\left[-x,\ y+\frac{1}{2},\ 0\right]$	`[3,22]`
4	$\left[-x,\ \frac{1}{2}-y,\ 0\right]$	`[4,21]`
5	$[-x,\ -y,\ 0]$	`[5,20]`
6	$[-x,\ y,\ 0]$	`[6,19]`
7	$\left[x,\ \frac{1}{2}-y,\ 0\right]$	`[7,18]`
8	$\left[x,\ y+\frac{1}{2},\ 0\right]$	`[8,17]`
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[9,32]`
10	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[10,31]`
11	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}\right]$	`[11,30]`
12	$\left[\frac{1}{2}-x,\ -y,\ \frac{1}{2}\right]$	`[12,29]`
13	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[13,28]`
14	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[14,27]`
15	$\left[x+\frac{1}{2},\ -y,\ \frac{1}{2}\right]$	`[15,26]`
16	$\left[x+\frac{1}{2},\ y,\ \frac{1}{2}\right]$	`[16,25]`

Table 15: Wyckoff site: `16o`, site symmetry: `m..`

No.	position	mapping
1	$[0,\ y,\ z]$	`[1,6]`
2	$[0,\ -y,\ -z]$	`[2,5]`
3	$\left[0,\ y+\frac{1}{2},\ -z\right]$	`[3,8]`
4	$\left[0,\ \frac{1}{2}-y,\ z\right]$	`[4,7]`

continued ...

Table 15

No.	position	mapping
5	$\left[\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[9,14]
6	$\left[\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[10,13]
7	$\left[\frac{1}{2},\, y,\, \frac{1}{2}-z\right]$	[11,16]
8	$\left[\frac{1}{2},\, -y,\, z+\frac{1}{2}\right]$	[12,15]
9	$\left[0,\, y+\frac{1}{2},\, z\right]$	[17,22]
10	$\left[0,\, \frac{1}{2}-y,\, -z\right]$	[18,21]
11	$[0,\, y,\, -z]$	[19,24]
12	$[0,\, -y,\, z]$	[20,23]
13	$\left[\frac{1}{2},\, y,\, z+\frac{1}{2}\right]$	[25,30]
14	$\left[\frac{1}{2},\, -y,\, \frac{1}{2}-z\right]$	[26,29]
15	$\left[\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[27,32]
16	$\left[\frac{1}{2},\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[28,31]

Table 16: Wyckoff site: 16p, site symmetry: .m.

No.	position	mapping
1	$\left[x,\, \frac{3}{4},\, z\right]$	[1,7]
2	$\left[x,\, \frac{1}{4},\, -z\right]$	[2,8]
3	$\left[-x,\, \frac{1}{4},\, -z\right]$	[3,5]
4	$\left[-x,\, \frac{3}{4},\, z\right]$	[4,6]
5	$\left[x+\frac{1}{2},\, \frac{1}{4},\, z+\frac{1}{2}\right]$	[9,15]
6	$\left[x+\frac{1}{2},\, \frac{3}{4},\, \frac{1}{2}-z\right]$	[10,16]
7	$\left[\frac{1}{2}-x,\, \frac{3}{4},\, \frac{1}{2}-z\right]$	[11,13]
8	$\left[\frac{1}{2}-x,\, \frac{1}{4},\, z+\frac{1}{2}\right]$	[12,14]
9	$\left[x,\, \frac{1}{4},\, z\right]$	[17,23]
10	$\left[x,\, \frac{3}{4},\, -z\right]$	[18,24]
11	$\left[-x,\, \frac{3}{4},\, -z\right]$	[19,21]
12	$\left[-x,\, \frac{1}{4},\, z\right]$	[20,22]
13	$\left[x+\frac{1}{2},\, \frac{3}{4},\, z+\frac{1}{2}\right]$	[25,31]
14	$\left[x+\frac{1}{2},\, \frac{1}{4},\, \frac{1}{2}-z\right]$	[26,32]
15	$\left[\frac{1}{2}-x,\, \frac{1}{4},\, \frac{1}{2}-z\right]$	[27,29]
16	$\left[\frac{1}{2}-x,\, \frac{3}{4},\, z+\frac{1}{2}\right]$	[28,30]

Table 17: Wyckoff site: 16q, site symmetry: .m'.

No.	position	mapping
1	$[x,\, 0,\, z]$	[1,23]
2	$[x,\, 0,\, -z]$	[2,24]
3	$\left[-x,\, \frac{1}{2},\, -z\right]$	[3,21]
4	$\left[-x,\, \frac{1}{2},\, z\right]$	[4,22]
5	$[-x,\, 0,\, -z]$	[5,19]
6	$[-x,\, 0,\, z]$	[6,20]
7	$\left[x,\, \frac{1}{2},\, z\right]$	[7,17]
8	$\left[x,\, \frac{1}{2},\, -z\right]$	[8,18]

continued ...

Table 17

No.	position	mapping
9	$\left[x+\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[9,31]
10	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[10,32]
11	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}-z\right]$	[11,29]
12	$\left[\frac{1}{2}-x,\ 0,\ z+\frac{1}{2}\right]$	[12,30]
13	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[13,27]
14	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[14,28]
15	$\left[x+\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	[15,25]
16	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	[16,26]

Table 18: Wyckoff site: 32r, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[x,\ -y,\ -z]$	[2]
3	$\left[-x,\ y+\frac{1}{2},\ -z\right]$	[3]
4	$\left[-x,\ \frac{1}{2}-y,\ z\right]$	[4]
5	$[-x,\ -y,\ -z]$	[5]
6	$[-x,\ y,\ z]$	[6]
7	$\left[x,\ \frac{1}{2}-y,\ z\right]$	[7]
8	$\left[x,\ y+\frac{1}{2},\ -z\right]$	[8]
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[9]
10	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[10]
11	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}-z\right]$	[11]
12	$\left[\frac{1}{2}-x,\ -y,\ z+\frac{1}{2}\right]$	[12]
13	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[13]
14	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[14]
15	$\left[x+\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$	[15]
16	$\left[x+\frac{1}{2},\ y,\ \frac{1}{2}-z\right]$	[16]
17	$\left[x,\ y+\frac{1}{2},\ z\right]$	[17]
18	$\left[x,\ \frac{1}{2}-y,\ -z\right]$	[18]
19	$[-x,\ y,\ -z]$	[19]
20	$[-x,\ -y,\ z]$	[20]
21	$\left[-x,\ \frac{1}{2}-y,\ -z\right]$	[21]
22	$\left[-x,\ y+\frac{1}{2},\ z\right]$	[22]
23	$[x,\ -y,\ z]$	[23]
24	$[x,\ y,\ -z]$	[24]
25	$\left[x+\frac{1}{2},\ y,\ z+\frac{1}{2}\right]$	[25]
26	$\left[x+\frac{1}{2},\ -y,\ \frac{1}{2}-z\right]$	[26]
27	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[27]
28	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[28]
29	$\left[\frac{1}{2}-x,\ -y,\ \frac{1}{2}-z\right]$	[29]
30	$\left[\frac{1}{2}-x,\ y,\ z+\frac{1}{2}\right]$	[30]
31	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[31]
32	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[32]